Shiyu Chen 陈时宇

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Research interests: autonomous exploration & active perception · semantic mapping & LiDAR–camera fusion · next-best-view planning · VLM-in-the-loop industrial inspection

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) — Aug 2025 – May 2027 (expected)
Computational Engineering & Robotics Lab (CERLAB), advisor Prof. Kenji Shimada — Pittsburgh, PA

University of Cincinnati & Chongqing University — B.S. / B.Eng. in Mechanical Engineering (joint program) — 2020 – 2025
Robotics Minor (UC) · GPA 3.71/4.0 (UC), 3.58/4.0 (CQU) · Dean's List ×4 — Cincinnati, OH / Chongqing, CN

RESEARCH EXPERIENCE

Graduate Research Assistant, CERLAB, Carnegie Mellon University (CMU × YKK project) — Oct 2025 – Present

5-DoF semantic exploration with VLM-guided active inspection co-first author · under review at ICRA 2027

- Proposed **5-DoF viewpoint planning** for Spot: body pitch/roll on top of planar motion reveals surfaces a level camera misses.
- Built 3D semantic mapping: per-instance segmentation masks back-projected into 3D object boxes in the occupancy map.
- Independently ran the full YOLOv8-seg pipeline: data collection, SAM2-assisted labeling, training and tuning, deployment.
- Viewpoint sampler scores body postures by expected coverage gain; aim-aligned execution avoids needless reorientation.
- Added **VLM-assisted viewpoint pruning** from observation history and a bird's-eye-view map, with a geometric fallback.
- Built the Isaac Sim benchmark: three industrial scenes, ground-truth surfaces, plug-in planners, Dockerized; planner in C++/ROS.
- **+8–10 pp coverage** and **17–32% less exploration time** vs. the planar 3-DoF baseline; highest mean object-coverage AUC.
- **53–73% fewer postures** than competing 5-DoF methods; demonstrated on a real Spot in a machine shop (five target machines).

Real-time LiDAR–camera semantic registration on a legged platform

- Calibrated the intrinsics of two cameras with unknown optical models (Spot CAM, Insta360 X4) in OpenCV.
- **Targetless registration** of Ouster OS1 point clouds and Insta360 X4 panoramas: extrinsics and transform jointly optimized.
- **10 FPS** at full LiDAR rate, **<100 ms** latency, **<10 px** reprojection error on a walking Boston Dynamics Spot.
- Iterated a 3D-printed rigid sensor mount until registration stayed stable under gait vibration.

3D reconstruction of shoe interiors for footwear fit Sep 2026 – Present · sole researcher

- Geometry-only 3D reconstruction of the inside of a shoe for fit assessment (e.g., diabetic feet); scope extended to garments.
- Route A: slender stereo **endoscopic probe**, two side-looking global-shutter cameras (34 mm baseline, 148° FOV), LED ring.
- Route B: cone-mirror laser ring and one wide-angle camera (**laser triangulation**); both heads share one rail carriage.
- Error budget: ≈0.3 mm depth error at 100 mm standoff from the 34 mm baseline; sharp range verified against lens depth of field.
- Designed the 3D-printed manual rail fixture (push/pull + roll) and a test plan that uses the CAD model as ground truth.

Course Project, Robust LiDAR Odometry with Vehicle-Kinematic Inequality Constraints — Sep – Dec 2025
CMU 24-785 Engineering Optimization, team of 3 — Pittsburgh, PA

- Converted KITTI / HeLiPR / AutoMine data (LiDAR, IMU, ground truth) to ROS 2; steering bounds from Ackermann geometry.
- Point-to-plane ICP with **bicycle-model inequality constraints** on yaw vs. forward motion, solved as active-set QPs in C++.
- Drift (RMSE / length) fell from **1.7% to 1.4%** on a 6.3 km mine route and from **1.9% to 1.4%** on a 19 km HeLiPR urban route.

Undergraduate Researcher, Cable-Driven Soft Robotic Arm for Minimally Invasive Surgery — Sep 2024 – May 2025
University of Cincinnati, advisor Prof. Janet (Jiangxia) Dong — Cincinnati, OH

- Designed and prototyped a wire-actuated continuum arm with 0.27 mm/step resolution and 230° bending range.
- Modeled nonlinear actuation; led hardware–control integration; co-author on the *ASME J. Medical Diagnostics* paper (2026).

Undergraduate Researcher, Deep-Learning Surface-Defect Segmentation — Jan – Apr 2022
Chongqing University, advisor Prof. Bo Yang — Chongqing, CN

- Semantic segmentation of aluminium surface defects (SegNet + ResNet encoder, 2,658 images), **97.64%** accuracy.
- Built an EfficientDet detection baseline first; collected 1,125 carbon-fibre vibration-source samples.

PUBLICATIONS & PATENTS

X. Zhan*, **S. Chen***, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," arXiv:2609.19460, Sep 2026. Under review at ICRA 2027. *Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME Journal of Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *Int. Conf. on Image Processing, Computer Vision and Machine Learning (ICICML)*, Chengdu, 2023.

Granted invention patents (China, co-inventor): CN117975197B · CN117218334B · CN117218393B · CN115588114B, on machine vision for steelmaking process monitoring (steel-stream landing point, slag splashing, slag discharge).

INDUSTRY EXPERIENCE

Computer Vision Engineer (Intern), CISDI Information Technology — May 2022 – Dec 2023 (3 co-op terms) · Chongqing, China

- Developed belt-conveyor inspection algorithms for steel plants, mainly semantic segmentation (UNet family, Transformer-based).
- Proposed a GAN that corrects predicted segmentation masks, improving material-state accuracy; compensated lens distortion.
- Proposed a Wasserstein-distance test for the particle-size algorithm, replacing manual comparison.
- Independently built a YOLOv5s object detector that supplies control signals for automatic converter tapping.
- Sped up a detector's TensorRT inference by **~40%**; ported an algorithm to C++ for DeepStream on Jetson Xavier NX / AGX Orin.
- Ported two inspection pipelines to Huawei Ascend (Atlas 200 DK / MindSpore).
- Built a team-wide annotation tool (own classification and video-frame modules); led 5 interns labeling ~100k images.
- Co-inventor on **4 granted computer-vision invention patents**; project featured on China Central Television.

TECHNICAL SKILLS

Programming	C++ (proficient), Python (proficient), MATLAB, C
Robotics	ROS 1 & 2, Isaac Sim, Spot SDK, SLAM & LiDAR odometry (ICP), LiDAR–camera calibration, stereo vision, frontier exploration, next-best-view / active perception, path planning (TSP/LKH), QP optimization
Perception / ML	PyTorch, TensorRT, OpenCV, YOLO (v5 / v8-seg), SAM2, VLM / VLA, GANs, point-cloud processing
Hardware / Tools	Jetson Xavier NX / AGX Orin, Ouster LiDAR, Linux, Docker, Git, SolidWorks, COMSOL, 3D printing

HONORS & TEACHING

- National Bronze Prize, 8th China International "Internet+" Innovation & Entrepreneurship Competition, 2023.
- Teaching Assistant — Engineering Design Thinking II (UC, 2025) and C++ Programming (CQU, 2024).